

# Limited Conservation Benefits From the First Marine Protected Area in the High Seas

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## Abstract

Can Large-Scale Marine Protected Areas be effective on the high seas? We address this question by evaluating the world's first large-scale spatial closure implemented in areas beyond national jurisdiction. In the Western and Central Pacific Ocean, two large high-seas pockets were closed to industrial purse-seine fishing in 2010 to reduce bycatch of juvenile bigeye tuna, but industrial longline fishing was allowed to continue. Nearly two decades later, the empirical consequences of this policy change remain unknown. We use publicly available data to evaluate this policy and find that the closure significantly reduced purse-seine fishing within the high-seas pockets and that longline vessels operating within the closed areas experienced a 19.6% increase in the number of bigeye tuna caught per 1000 hooks. We find no evidence that the closure generated a spillover benefit to purse-seine vessels fishing near the boundaries of the closed areas.

**Significance statement.** The High Seas Treaty entered into force in January 2026, yet no empirical evidence exists on whether Marine Protected Areas can deliver conservation benefits beyond national jurisdiction. We provide this evidence by evaluating the world's first large-scale high-seas closure—two vast Pacific fishing grounds totaling over one million km<sup>2</sup> closed to industrial purse seining since 2010. Analyzing two decades of spatially explicit fishing data, we show that market-based enforcement successfully eliminated industrial fishing from international waters. However, the resulting conservation benefits were far more modest than those documented for comparable closures within national waters, and no spillover reached adjacent fisheries. As nations begin designating high-seas MPAs under the new treaty, our findings provide the only empirical benchmark for what these protections may—and may not—achieve.

# 1 Introduction

The entry into force of the UN High Seas Treaty (BBNJ Agreement) on January 17, 2026, officially established the legal framework necessary to create and manage Marine Protected Areas (MPAs) on the high seas. By definition, high seas MPAs will protect pelagic ecosystems. In this way, they will share some similarities with Large-Scale MPAs (LSMPAs; areas larger than 100,000 km<sup>2</sup> [1, 2]). By virtue of the habitats they protect, LSMPAs most often restrict fishing effort of pelagic fisheries, such as those targeting tuna, billfish, and sharks [3]. This is also likely to be one of the primary impacts of high seas MPAs.

Previous work has found that tuna fisheries operating near LSMPA boundaries may see increases in their catch rates [4, 5, 6]. For example, Hilborn et al. [6] finds that the CPUE of bigeye tuna has increased by 13% outside the boundaries of Papahānaumokuākea National Marine Sanctuary in Hawai‘i. This spillover benefit may prompt nations to think strategically when designing LSMPAs on the high seas: nations may implement LSMPAs immediately outside their Exclusive Economic Zones (EEZs) hoping to boost catch rates in their waters [7, 8]. But would this work?

There are subtle but important differences between Large-Scale MPAs inside EEZs and Large-Scale MPAs on the high seas. One important difference is enforcement and compliance. Not all nations have ratified the UN Convention on the Law of the Sea and it is still unclear who would be responsible for enforcing and potentially confronting illegal fishing in future high seas MPAs. A second important difference is habitat and environmental conditions. High seas MPAs will all be located at least 200 nautical miles away from land, in some of the deepest and most remote marine ecosystems in the world. Many of the processes that influence pelagic species around islands and seamounts are either absent or very different on the high seas [9, 10, 11, 12]. Therefore, results from LSMPAs within national jurisdictions may not extrapolate to similar sized MPAs on the high seas [13]. Here, we fill this crucial gap in policy-relevant knowledge by studying the first known case of a large-scale area-based marine conservation on the high seas.

We focus on the Western and Central Pacific Ocean, where industrial tuna fishing vessels use two main gears to target different tuna species. The longline fleet primarily catches albacore (*Thunnus alalunga*), bigeye (*Thunnus obesus*), and yellowfin tuna (*Thunnus albacares*). The purse seine fleet mainly catches skipjack (*Katsuwonus pelamis*) and yellowfin tuna, with some bycatch of juvenile

bigeye tuna associated with the use of drifting Fish Aggregating Devices (dFADs) [14, 15, 16].

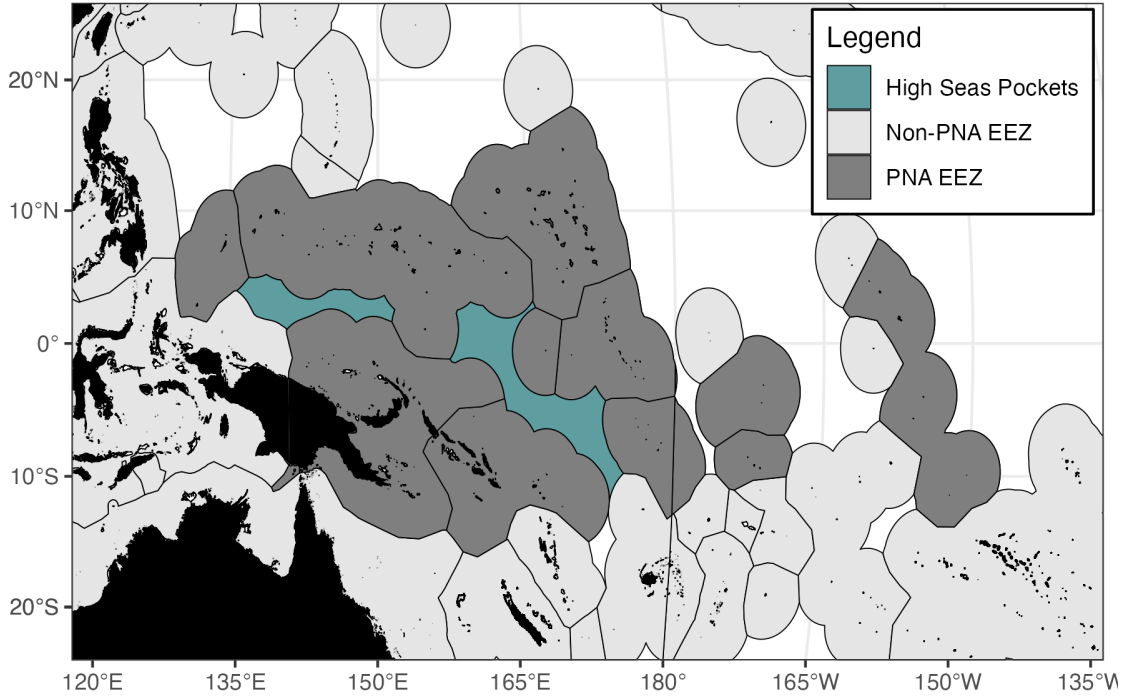


Figure 1: Map of the Western and Central Pacific Ocean. Light gray polygons show the Exclusive Economic Zones of non-PNA nations. Dark gray polygons highlight the Exclusive Economic Zones of the eight Parties to the Nauru Agreement nations and Tokelau. The two teal polygons show the High Seas Pockets—the “donut holes” formed by the surrounding Exclusive Economic Zones—where purse seine fishing has been prohibited since 2010.

The Parties to the Nauru Agreement (PNA<sup>1</sup>)—a group of eight nations in the region—and Tokelau jointly manage tuna fisheries in their waters using a market-based approach known as the Vessel Day Scheme (VDS; [17, 18, 19]). Bycatch of juvenile tuna by the purse seine fleet presented a major problem to the longline fleet [14, 15, 16], particularly in the two high seas pockets formed between the PNA’s Exclusive Economic Zones (EEZs; Figure 1). When the spatial overlap between target and bycatch species is severe, MPAs may be a useful tool to protect the “weak stock” [20], here juvenile bigeye. In an effort to reduce the purse-seine fleet’s bycatch of juvenile bigeye tuna—an externality affecting longline vessels targeting adult bigeye tuna—purse seine fishing was prohibited in the high seas pockets starting in 2010 [21, 22]. These pockets are

<sup>1</sup>PNA members include: Federated States of Micronesia, Kiribati, Nauru, Palau, Papua New Guinea, Republic of the Marshall Islands, Solomon Islands, and Tuvalu.

roughly 360,000 km<sup>2</sup> and 780,000 km<sup>2</sup>, so they are very much in line with the size of LSMPAs in the region, like the Palau National Marine Sanctuary (475,077 km<sup>2</sup>) or the now degazetted Phoenix Island Protected Area (408,422 km<sup>2</sup>). In closing the high seas pockets to industrial tuna purse seine fisheries, this intervention effectively created the first partially protected *de facto* large-scale MPA in areas beyond national jurisdiction. How was the prohibition on purse seine fishing achieved before the UN High Seas Treaty even existed? The PNA used a carrot-and-stick approach: vessels that adhered to the closure were allowed to fish in the lucrative waters of the PNA countries under the VDS program, vessels that ignored the purse seine ban were excluded from the VDS program<sup>2</sup>.

There are only a few examples of high seas closures implemented before the adoption of the high seas treaty, and their potential effectiveness remains contested or unevaluated [13]. Previous work has claimed that the benefits of high seas closures to tuna populations would be limited. For example, Sibert et al. [34] used hind-cast simulations to estimate the impact that the PNA's high seas pockets closures would have on bigeye and skipjack tuna populations, finding that they would have negligible stock-wide effects (0.1-0.8% increase) but potential for spillover (10-25% localized increase), depending on model assumptions. However, nearly two decades later, the actual effectiveness of this spatial intervention remains untested. More importantly, recent empirical work on LSMPAs within national jurisdictions has found that catch rates tend to increase in areas near LSMPA boundaries and abundances may increase locally [4, 5, 6].

Our objective is to test for the effectiveness of this unique policy intervention: the first large-scale spatial closure on the high seas. We compile a multi-decadal (2000-2020) and spatially explicit dataset on catch and effort by purse seine and longline fisheries using publicly available information from the Western and Central Pacific Fisheries Commission (WCPFC; see section 3). We present our analysis in three parts. First, we ask whether the intervention actually reduced purse seine fishing in the high seas pockets. Second, we test whether the reduction of purse seine fishing effort benefited the longline fleet that still fishes in the pockets (as originally hoped for). Third, we test whether the high seas pockets, acting as a *de facto* marine protected area, provided a spillover

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<sup>2</sup>The WCPFC introduced a small exception for the Philippines artisanal purse seine fleet, which allowed limited fishing by a small number of vessels in the north-western pocket, as stated in a series of conservation and management measures Western and Central Pacific Fisheries Commission [23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33]. Additional analyses in our supplementary materials address this.

benefit to tuna fisheries fishing outside but near the high seas boundaries (an unintended benefit observed in other settings). We answer these questions following a “pre-registered” study design available in the Open Science Framework platform [35]. As we will show, the intervention was successful at eliminating purse seine fishing from the high seas pockets, but we only observe weak evidence that this led to a benefit to the longline fleet, and no evidence of a spillover benefit. We believe our findings can help inform the design of, and manage expectations around, proposed LSMPAs on the high seas.

## 2 Results and Discussion

### 2.1 Reducing fishing effort by purse seiners

The closures sought to eliminate purse seine fishing effort from two high seas pockets. Consequently, we begin by testing for changes in purse seine fishing effort in the high seas pockets. A visual inspection of trends in the data shows that, in the 10 years leading to the closure, total annual fishing effort by purse seiners fishing in the high seas pockets was around  $2,187 \pm 770$  days (mean  $\pm$  standard deviation) or  $1,766 \pm 634$  sets (Figure 2A-B). After the closure, fishing effort in the pockets dropped to zero starting in 2010, except for a few events recorded in 2013 and 2019.

To conclusively claim that vessels largely vacated the area, we must compare trends in fishing effort inside the high seas pockets to trends in fishing effort in nearby comparable areas. These “control” areas must be subject to the same environmental conditions as well as fishery management rules, thus providing a counterfactual of expected fishing effort if the closure had not been implemented. With this in mind, we compare all  $1^\circ \times 1^\circ$  grid cells in the high seas pockets ( $N = 40$ ) to other high seas grid cells that fall under the WCPFC convention area and lie between  $20^\circ\text{N}$  and  $20^\circ\text{S}$  ( $N = 151$ ; see Figure A3). We estimate a two-way fixed effects regression model that accounts for differences across grid cells and years to test for a causal effect of the closure (see Methods in section 3.4.1 for more details).

In line with the patterns in Figure 2, we find that the closure dramatically reduced fishing effort in the high seas pockets, with an estimated annual reduction of 59.47 days or 50.93 sets per grid cell per year ( $p < 0.01$ ), relative to trends observed for other high-seas areas in the region (Table 1). We also estimate a model with dynamic treatment effects that measures the change in fishing effort

by year. These event-study regressions reveal two important findings. First, that treatment effect estimates before the treatment are not statistically different from zero, indicating that effort in our treated and control areas followed similar trends before the closure. And second, that all years since 2010 had significantly less fishing effort than what would have been expected, had the closure never been implemented ( $p < 0.01$ ; Figure 2C-D). Results for other specifications are provided in Supplementary Materials section B.

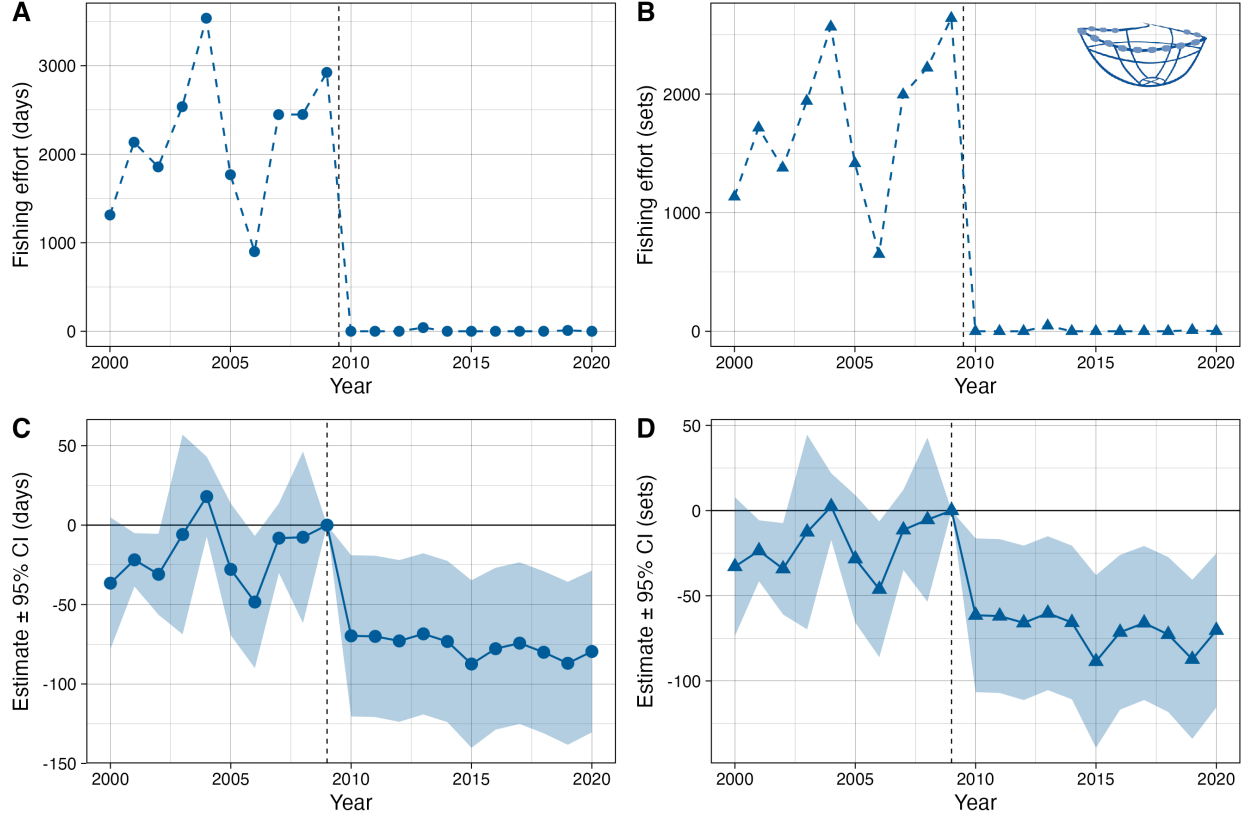


Figure 2: Change in purse seine fishing effort in the high seas pockets (combined). Panels A and B show annual total fishing effort measured in days and number of sets, respectively, across all grid cells in the high seas pockets. Panels C and D show event-study plots for the change in fishing effort in the high seas pockets, relative to fishing effort in comparable high seas regions in the WCPFC convention area.

Table 1: Coefficient estimates for change in fishing effort inside the high seas pockets after the closure, relative to changes in fishing effort observed for other comparable high seas areas in the WCPFC convention area.

|                 | Effort (days)          | Effort (sets)          |
|-----------------|------------------------|------------------------|
| Post x Treated  | −59.470***<br>(13.762) | −50.933***<br>(10.607) |
| $\bar{Y}_{pre}$ | 54.670                 | 44.145                 |
| Num.Obs.        | 4011                   | 4011                   |
| R2 Adj.         | 0.388                  | 0.301                  |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors using a 220km radius.  $\bar{Y}_{pre}$  indicates the mean of each outcome variable over the pre-closure period.

## 2.2 Benefits to the longline fishery

In the 10 years before closure (2000-2009), tuna purse seines caught an average of 3,200 metric tons (mt) of bigeye tuna<sup>3</sup>, 48,500 mt of skipjack tuna, and 13,800 mt of yellowfin tuna per year inside the two high seas pockets (Figure A1). Our results show that purse seine fishing effort was significantly reduced in the high seas pockets in 2010. It follows, then, that so was catch by purse seiners. Recall that the longline fleet has always been allowed to operate in the high seas pockets, so now we ask: Did reducing purse seine fishing mortality in the high seas pocket benefit the longline fleet?

We focus on bigeye tuna because the closure was motivated by high bycatch rates of juvenile bigeye in the purse seine fishery ([14, 34, 15, 16]; see Supplementary Materials section B.2 for results on all other species). We measure benefits to the longline fleet using catch-per-unit-effort (CPUE) of bigeye tuna. Specifically, we measure CPUE in number of fish for every thousand hooks and metric tons for every thousand hooks, which can respectively be understood as imperfect proxies for local abundance and biomass [36]. Time series of both measures of bigeye CPUE from the longline fleet operating in the high seas pockets and control regions are shown in Figure 3A-B.

<sup>3</sup>To put 3,200 metric tons in context, the Total Allowable Catch for the US-flagged longline fleet for the entire Western and Central Pacific in 2024 was 3,554 metric tons.

Both measures show a steady decline in CPUE during the ten years before the exclusion of purse seiners. After the closure, the decreasing trend slows down and CPUE shows relative stability, particularly when measured in number of fish per thousand hooks (Figure 3A). Note, too, that there was a sharp increase in CPUE starting around 3 years after the closure, peaking at around 5 years, and then returning to a steady level seven years later.

As before, a formal test requires that we compare trends in CPUE in the high seas pockets to an appropriate control region. In this second test, we retain all  $5^\circ \times 5^\circ$  grid cells that fall in the high seas pockets (our “treated” cells;  $N = 16$ ) and all high seas grid cells in the WCPFC convention area that lie between  $20^\circ\text{N}$  and  $20^\circ\text{S}$  ( $N = 69$ ; Figure A4). We then estimate annual changes in bigeye CPUE in high seas pocket grid cells, relative to changes observed in our counterfactual grid cells (see Methods section 3.4.2). We detect a modest increase in bigeye CPUE of 0.69 fish per thousand hooks ( $p < 0.1$ ), and no change in CPUE when measured in tons per thousand hooks (Table 2). In terms of percentage changes, the increase in bigeye CPUE represents a 19.6% increase above the pre-treatment mean. The contrasting results across the two measures of CPUE suggest that fishers may have caught more fish per unit effort between 2010 and 2020, but that these fish were, on average, smaller. It is also very likely that the weight measure has more measurement error (it is easy, and standard practice, to record every bigeye caught, but not every fish is weighed individually and weight may be approximations made by the crew or extrapolated at the point of landing). Measurement error will bias coefficient estimates towards zero, in line with what we observe here.

Our event-study plots echo these aggregate results: the annual number of bigeye tuna caught per every thousand hooks exhibits a slightly positive trend since 2010, but we observe no change in bigeye CPUE when measured in metric tons per thousand hooks (Figure 3C-D). The lack of a monotonic increase in bigeye CPUE could be caused by a subsequent increase in fishing effort following the localized and temporary recovery of the stock. Similar tests for other species caught by the longline fleet and other model specifications are presented in the Supplementary Materials. One result to note is an estimated 38.9% increase in yellowfin tuna CPUE inside the high seas pockets when estimated using log-transformed numbers per 1,000 hooks (coefficient = 0.329, so  $(e^{0.329} - 1) \times 100 = 38.95$  Table B6). Results for other specifications and species are provided in Supplementary Materials section B.2. Tests for each high seas pocket separately are presented in



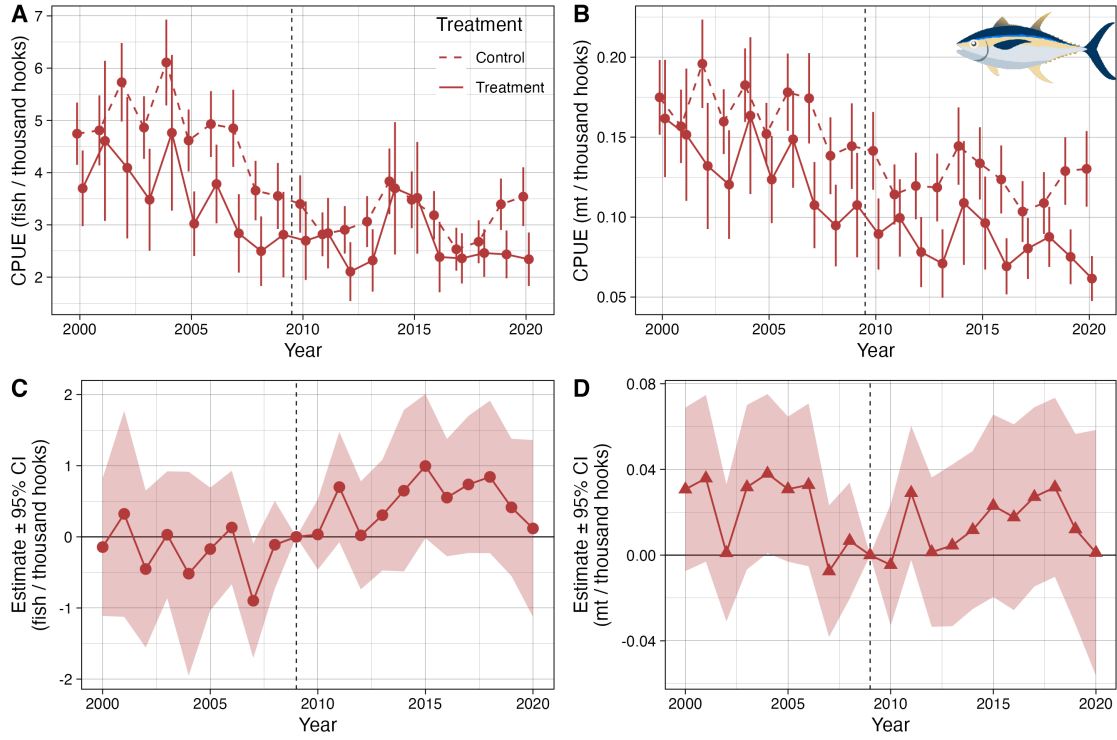


Figure 3: Change in catch-per-unit-effort (CPUE) of bigeye tuna in the longline fleet. Panels A and B show time series of mean CPUE in number of fish per 1000 hooks and metric tons per 1000 hooks, respectively, across all grid cells in the high seas pockets (solid lines) and other high seas grid cells (dashed lines). Panels C and D show event-study plots for the change in bigeye tuna CPUE in the high seas pockets, relative to CPUE in other tropical (20°S to 20°N) high seas regions in the WCPFC convention area.

Table 2: Coefficient estimates for change in bigeye tuna CPUE caught by the longline fleet in the high seas pockets after the closure, relative to changes in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area.

|                 | fish / 1000 hooks | mt / 1000 hooks |
|-----------------|-------------------|-----------------|
| Post x Treated  | 0.697*            | -0.003          |
|                 | (0.363)           | (0.011)         |
| $\bar{Y}_{pre}$ | 3.548             | 0.131           |
| Num.Obs.        | 1631              | 1631            |
| R2 Adj.         | 0.798             | 0.829           |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius.  $\bar{Y}_{pre}$  indicates the mean of each outcome variable over the pre-closure period.

### 2.3 Spillover benefits to the purse seine fishery

Skipjack tuna in the open ocean are mainly targeted by the purse seine fleet, with the longline fishery representing less than 1% of total skipjack caught in the WCPFC convention area [37]. When the high seas pockets were closed to purse seine fishing, the region essentially became a *de facto* no-take zone for skipjack tuna. Previous work has shown that the CPUE of tuna purse seiners often, but not always, increases near large-scale MPAs after their implementation [5]. Therefore, we now ask whether the high seas pockets, by substantially reducing localized fishing mortality on skipjack, also produced spillover benefits to tuna purse seine fisheries operating near the borders of the pockets. A direct examination of a time series of skipjack tuna CPUE in Figure 4 shows two contrasting trends. When measured in metric tons *per day*, CPUE within 100 nautical miles of the high seas pockets appears to increase after 2010, as compared to pre-closure years (Figure 4A). We observe the opposite when we measure CPUE in metric tons *per set*, which shows lower CPUE for most years after the closure (Figure 4B).

Following previous work, we formally test for changes in skipjack CPUE near the high seas pocket boundaries (within 100 nm;  $N = 226$ ), relative to changes observed far from the boundary (100–200 nm;  $N = 184$ ) [4, 5, 6]. We only retain grid cells that fall within PNA countries' waters because purse seine fishing is strictly managed under the vessel-day scheme in these waters (com-

pared to the high seas) [17, 5]. Using a standard Before-After-Near-Far design, we fail to detect changes in CPUE that would be indicative of a spillover effect, with estimated effects of -0.49 tons per day and -0.86 tons per set ( $p = 0.45$  and  $p = 0.29$ , respectively; Table 3). Our event-study estimates are shown in panels C and D of Figure 4. These highlight that CPUE near the high seas pockets is not different from CPUE observed 100-200nm from the pockets. Overall, there is no evidence to indicate that closing the high seas pockets produced spillover benefits to adjacent purse seine vessels targeting skipjack tuna. Similar results for other species and additional model specifications are presented in the Supplementary Materials section B.3. Tests for each high seas pocket separately are presented in section C.1.

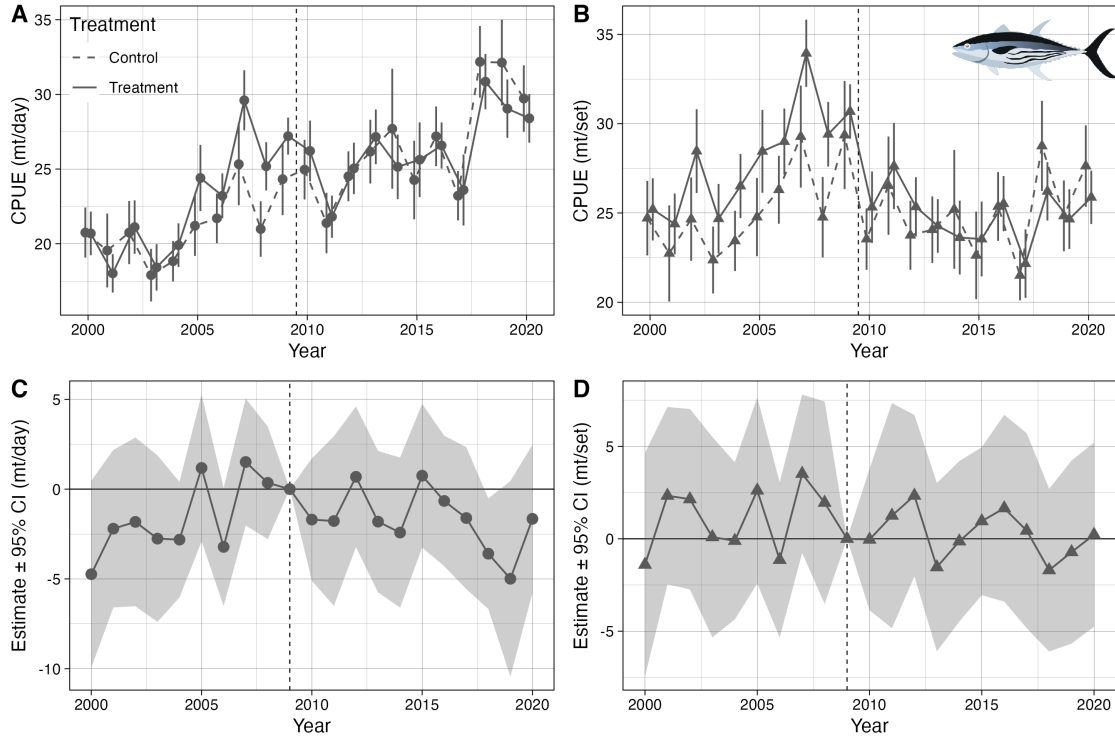


Figure 4: Change in catch-per-unit-effort (CPUE) of skipjack tuna in the purse seine fleet. Panels A and B show time series of mean CPUE in tons per day and tons per set, respectively, across all grid cells within 100 nm of the high seas pocket (solid) and those far away (100–200 nm, dashed). Panels C and D show coefficient estimates for the change in skipjack tuna CPUE near the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Both the near and far purse seine samples are restricted to grid cells within PNA member Exclusive Economic Zones.

Table 3: Coefficient estimates for change in skipjack tuna CPUE in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles and inside PNA nation’s Exclusive Economic Zones.

|                 | CPUE (mt/day)     | CPUE (mt/set)     |
|-----------------|-------------------|-------------------|
| Post x Near     | −0.496<br>(0.679) | −0.865<br>(0.818) |
| $\bar{Y}_{pre}$ | 22.946            | 28.182            |
| Num.Obs.        | 5960              | 5960              |
| R2 Adj.         | 0.333             | 0.245             |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.  $\bar{Y}_{pre}$  indicates the mean of each outcome variable over the pre-closure period.

## 2.4 Policy Implications for Large-Scale MPAs on the High Seas

We evaluated changes in fishing effort and catch-per-unit effort inside and near two high seas areas where a closure on purse seine fishing has been implemented since 2010. Our results show that purse seine fishing effort was significantly curtailed in the two high seas pockets. This intervention led to longliners catching 19.6% more bigeye tuna per unit effort momentarily. But, we do not observe evidence that the closure boosted catch rates of skipjack tuna for purse seine vessels fishing near the high seas pockets.

The success in eliminating fishing effort within the high seas pockets highlights the power of environmental markets. PNA waters contain some of the most productive skipjack tuna fishing grounds in the world and the majority of the Pacific resource is typically caught in these waters [17, 18]. This high productivity and strong market power gives the PNA members considerable bargaining power when negotiating access agreements for foreign fishing vessels. As we have shown, this allowed the PNA members to close a portion of the high seas to fishing, well before the High Seas Treaty was even ratified. The PNA countries essentially established a property right in an area with no formal property rights.

Even though the intervention successfully reduced purse seine fishing in the high seas pockets,

the evidence of benefits to the longline fleet is somewhat limited. We find that longliners catch more fish, but not more tons of fish, per unit effort. This is broadly aligned with initial expectations. Because purse seiners are no longer incidentally catching juvenile bigeye tuna, more of these fish are available for harvest by the longline fleet. It is plausible that there are cohorts of juvenile bigeye that evade purse seine mortality within the pockets and some survivors later show up on longline hooks. But continued or even higher longline fishing mortality within the pockets would reduce the probability of survival into larger size classes.

The weak and null effects we detect could be due to the resolution of the data. Longline data are reported on a  $5^\circ \times 5^\circ$  grid, which may be too coarse to detect the expected small effects [34]. Unfortunately, the WCPFC does not make higher resolution data publicly available but, in principle, these exist and would allow for a more granular test as done elsewhere [38, 5, 6]. Of course, another explanation is that the high seas closures are simply not providing any conservation benefits, suggesting that traditional fisheries management approaches such as regulating the use of drifting Fish Aggregating Devices across the region could have been more effective [34, 16].

We fail to detect any benefit to purse seine fisheries operating near the high seas pockets, even though the closure significantly reduced fishing mortality for skipjack tuna. This may be because longliners continued operating in the region and they occasionally catch skipjack tuna. This lingering source of fishing mortality means that skipjack are only partially protected, and previous work has shown that fully protected MPAs are the ones most likely to deliver conservation benefits [39].

Differences in habitat are another possible explanation for the null findings compared to MPAs situated inside of EEZs where spillover has been documented [4, 5, 6]. For example, previous work has shown that pelagic species tend to aggregate around shallow ( $< 150$  m deep) seamounts and islands [9, 10, 11, 12, 40]. Most Large-Scale MPAs within EEZs are placed around volcanic islands so it is not surprising that these MPAs contain multiple islands and shallow seamounts that produce the heterogeneous habitat that aggregates highly mobile pelagic species. The most comprehensive global dataset on seamount location and depth [41, 42] shows that the high seas pockets do not contain any shallow seamounts (Figure A2). This observation would lend support for calls for high seas MPAs to focus on these underwater features [43].

In conclusion, our results show that PNA members were able to use their bargaining power—

rooted in successful fisheries management—to significantly reduce fishing effort in the high seas pockets. This intervention led to the world’s first *de facto* large-scale MPA in areas beyond national jurisdiction. We find mixed evidence around the effects of this significant reduction in fishing effort, indicating that spillover benefits from high seas MPAs may be weaker than those observed to date around similar-sized MPAs within EEZs. Our results also show that lessons learned from LSMPAs implemented within national jurisdictions may not directly translate to those implemented in the high seas. They also highlight how features such as degree of protection, amount of fishing displaced, and presence of critical habitat may drive potential outcomes of future *de jure* MPAs in the high seas.

### 3 Methods

Prior to running any empirical analysis, we pre-registered our analysis plans using the AsPredicted template in the Open Science Framework and have subsequently followed those plans exactly. Pre-registration plans aim to prevent p-hacking, *ex post* hypothesizing, and discarding important null findings [44, 45]. The pre-registration plan states the hypotheses being tested, the outcome variables and units of measurement, and the process by which conditions (e.g. treated vs. control) are assigned. Our pre-registered plan is available at [35].

We set out to test three hypotheses: (1) the PNA’s closure of the high-seas pockets to purse seine fishing reduced purse seine fishing effort within the high seas pockets; (2) reducing purse seine fishing effort within the high seas pockets will increase catch-per-unit-effort of tuna, especially bigeye tuna, for the longline fleet operating within the high seas pockets; and (3), the closure will also increase catch-per-unit-effort of bigeye, skipjack, and yellowfin tuna for the purse seine and longline fleets operating in PNA waters within 100 nautical miles of the high seas pockets, relative to catch-per-unit-effort 100-200 nautical miles from the high seas pockets. The methods below describe the steps we take to build our panel dataset and test our hypotheses.

#### 3.1 Data sources

We primarily used publicly available data provided by the Western and Central Pacific Fisheries Commission (WCPFC). We downloaded data from the WCPFC Public Domain data por-

tal.<sup>4</sup> Specifically, the purse seine data came from the following file: “*Aggregated data, grouped by*  
*1°x1° latitude/longitude grids, YEAR and MONTH*” (WCPFC\_S.PUBLIC\_BY\_1x1\_MM5). And  
the longline data came from “*Aggregated data, grouped by 5°x5° latitude/longitude grids, YEAR*  
*and MONTH*” (WCPFC\_L.PUBLIC\_BY\_YR\_MON\_3). Both datasets were downloaded on Oct 31,  
2025.

## 3.2 Dataset construction

### 3.2.1 Outcomes of interest

Our two outcomes of interest are fishing effort and catch-per-unit-effort (CPUE). We measured  
fishing effort by the purse seine fleet in both days and sets. These are reported monthly as 1° by  
1° grid cell totals. We summed the monthly totals to obtain annual measures of effort by grid  
cell. We calculate CPUE per grid cell and year as the sum of the total catch divided by the total  
effort reported for that cell, adjusting for gear-specific differences. Longline CPUE is measured in  
number of fish per thousand hooks and tons per thousand hooks. For purse seine vessels, CPUE is  
measured in tons of fish per day and tons of fish per set.

## 3.3 Treatment and Control Groups

All regression-based hypothesis tests have time variables indicating for before or after the high seas  
closures (effective January 2010), as well as indicator variables for treatment and control groups.  
The definition of treatment and control groups varies by hypothesis test. When testing for changes  
in purse seine fishing effort, we define treated grid cells as those within the high seas pockets,  
and control grid cells as other high seas grid cells that fall under the WCPFC convention area  
and lie between 20°N and 20°S. When testing for changes in CPUE to the longline fleet operating  
in the high seas pockets, we consider grid cells inside high seas pockets as “treated” and other  
high seas grid cells that fall under the WCPFC convention area and lie between 20°N and 20°S as  
“control”. Finally, when testing for spillover benefits, we follow the same Before-After-Near-Far  
design used to test for spillover benefits from Large-Scale Marine Protected Areas in other settings  
[4, 5]. Specifically, grid cells within 100 nautical miles of the high seas pockets’ boundaries are

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<sup>4</sup>[www.wcpfc.int/wcpfc-public-domain-aggregated-catcheffort-data-download-page](http://www.wcpfc.int/wcpfc-public-domain-aggregated-catcheffort-data-download-page)

labeled as treated. Grid cells between 100 and 200 nautical miles distance from the boundaries are labeled as controls. For the purse seine fishery, we only retain grid cells that fall within PNA countries' EEZs to account for the sharp difference in management between vessels that operate under the vessel-day scheme *vs.* those fishing on the high seas [17, 18, 19].

### 3.4 Identification and Estimation

#### 3.4.1 Testing for changes in fishing effort inside the high seas pockets

Let  $E_{it}$  be purse seine effort (measured in sets or days) in pixel  $i$  at time  $t$ . We first test for changes in fishing effort in the high seas pockets with a simple linear regression that does not account for changes in region-wide fishing effort. The linear regression has the following form:

$$E_{it} = \sum_{t=-9, t \neq 0}^{10} \beta_t T_t + \mu_i + \epsilon_{it} \quad (1)$$

where  $T_t$  are indicators for periods relative to year of treatment, such that  $t = 0$  corresponds to the last year when purse seine fishing was allowed, 2009. Negative values of  $t$  are pre-closure years, and positive values of  $t$  are post-closure years. The parameter of interest is  $\beta_t$ , which is a vector of coefficients capturing changes in effort inside the high seas pocket relative to the last year when fishing was allowed (2009). Then,  $\mu_i$  are grid-cell fixed-effects and  $\epsilon_{it}$  is the error term, which accounts for spatio-temporal autocorrelation via Conley standard errors. These results are shown in Figure B1, which capture changes in effort relative to 2009. However, our main specification estimates changes across time relative to changes in effort in other high seas pixels:

$$E_{it} = \sum_{t \neq 0} \beta_t (T_t \times D_i) + \mu_i + \tau_t + \epsilon_{it}. \quad (2)$$

Here,  $D_i = 1$  indicates pixel  $i$  falls within the high seas closure and  $D_i = 0$  indicates other high seas cells not subject to the closure.  $\tau_t$  are fixed-effects by year and  $\mu_i$  is defined as above. This model allows us to account for region-wide trends in fishing effort, and under standard assumptions, allows us to derive a causal estimate of the intervention. The results are shown in Figure 2 Panels C and D.

Aggregate treatment effects were also estimated using the following regression models:  $E_{it} =$



$\beta\text{Post}_t + \mu_i + \epsilon_{it}$  and  $E_{it} = \beta\text{Post}_t \times D_i + \mu_i + \tau_t + \epsilon_{it}$ .  $\text{Post}_t$  is a dummy variable that takes the value of 1 or 0 if the observation comes from after or before the implementation period, respectively. Results are shown in Tables 1, B1, and B2.

Note that our pre-registration plan stated that we would also estimate models using log-transformed outcomes. However, due to the large number of zeroes in the data set, we instead use an inverse-hyperbolic sine transformation, considered a standard method for cases like this [46]. Event-study plots are shown in Figures B1 and B2, while aggregate results are shown in Tables B1 and B2.

### 3.4.2 Testing for changes in CPUE inside the high seas pockets

Let  $y_{it}$  denote longline catch-per-unit-effort (CPUE) in pixel  $i$  at time  $t$ . We estimate:

$$y_{it} = \sum_{t \neq 0} \beta_t (T_t \times D_i) + \mu_i + \tau_t + \epsilon_{it} \quad (3)$$

where all regressors have the same definitions as above. The coefficient of interest,  $\beta_t$ , captures the change in CPUE within the high seas pocket relative to changes observed in other high seas pixels where purse seine effort was not curtailed. A pre-post model with the form  $y_{it} = \beta\text{Post}_t \times D_i + \mu_i + \tau_t + \epsilon_{it}$  was estimated to test for aggregate effects. We also estimated log-linear versions of this model. Results for bigeye tuna are shown in Figure 3 Panels C and D and in Table 2. Results for other specifications and other species are shown in the Supplementary Materials section B in Tables B3–B6 and Figures B3–B6.

### 3.4.3 Testing for spillover benefits

Let  $y_{it}$  denote longline or purse seine catch-per-unit-effort (CPUE) in pixel  $i$  at time  $t$ . We estimate the following regression:

$$y_{it} = \sum_{t \neq 0} \beta_t (T_t \times D_i) + \mu_i + \tau_t + \epsilon_{it} \quad (4)$$

where the coefficient of interest,  $\beta_t$ , captures the dynamic treatment effect of the high seas closure on CPUE within 100 nautical miles of the high seas pockets, relative to changes observed in other

grid cells between 100 and 200 nautical miles from the high seas pockets. A pre-post model with the form  $y_{it} = \beta \text{Post}_t \times D_i + \mu_i + \tau_t + \epsilon_{it}$  was also estimated to test for aggregate effects along with log-linear specifications. Results for skipjack tuna in the purse seine fleet are shown in Figure 4 Panels C and D and in Table 3. Results for other specifications and species are shown in Supplementary Materials section B.3 in Tables B7–B14 and Figures B7–B14.

#### 3.4.4 Weights

By construction,  $y_{it} = \frac{\sum_{j \in (i,t)} \text{Catch}_j}{\sum_{j \in (i,t)} \text{Effort}_j}$ , so  $\text{Var}(y_{it}) \propto \frac{1}{\text{Effort}_{it}}$ . We therefore assume

$$\text{Var}(\epsilon_{it} \mid X) = \frac{\sigma^2}{\omega_{it}}, \quad \omega_{it} = \text{Effort}_{it},$$

and estimate all models using normalized measures via weighted least squares using effort as weights.

#### 3.4.5 Software and Reproducibility

All data cleaning and sample construction was performed with the tidyverse (v2.0.0), sf (1.0-21), and terra (1.8-60) packages using R (v4.5.1) and RStudio (2026.07.1 Build 147 “Pacific Dogwood” Wickham et al. [47], Pebesma [48], Pebesma and Bivand [49], Hijmans [50], R Core Team [51]). All models were fit using the fixest package (v 0.13.2, Bergé [52]). Regression tables were built using the modelsummary package (v1.3.0, Arel-Bundock [53]), and event-study plots were built with ggfixest (v0.4.0, McDermott [54]) and customized with the ggplot2 package (v4.0.2, [55]). All our data and code are publicly available in a GitHub repository posted at [https://github.com/jcvdav/PNA\\_HS\\_closure](https://github.com/jcvdav/PNA_HS_closure).

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# Appendix

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Supplementary Materials

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For

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**Limited Conservation Benefits From the First Marine Protected Area in the**

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**High Seas**

## A Supplementary Figures and Tables

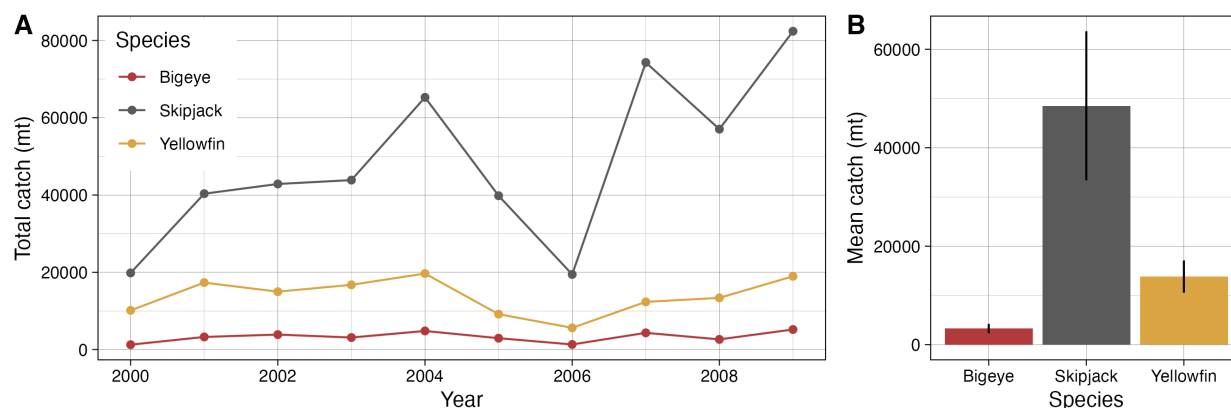


Figure A1: Historical (2000-2009) catch of bigeye, skipjack, and yellowfin tuna by tuna purse seiners fishing in the high seas pocket before its closure. Panel A shows a time series of catch by species, panel B shows mean ( $\pm$  SD) catch by species.

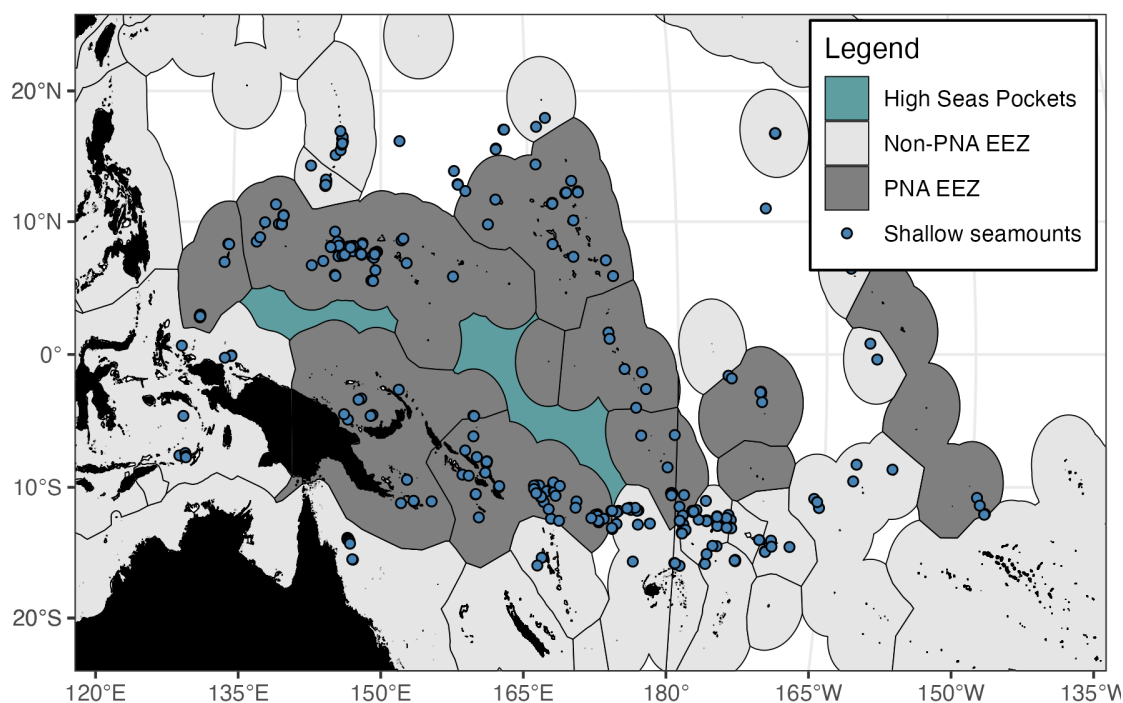


Figure A2: Map of shallow seamounts in the Western and Central Pacific. Blue points indicate the centroid of shallow seamounts ( $\leq 150$  m deep) as defined in [41, 42]. Light gray polygons show the Exclusive Economic Zones of all nations. Dark gray polygons highlight the Exclusive Economic Zones of PNA nations. The two teal polygons show the High Seas Pockets closed to purse seine fishing starting in 2010.

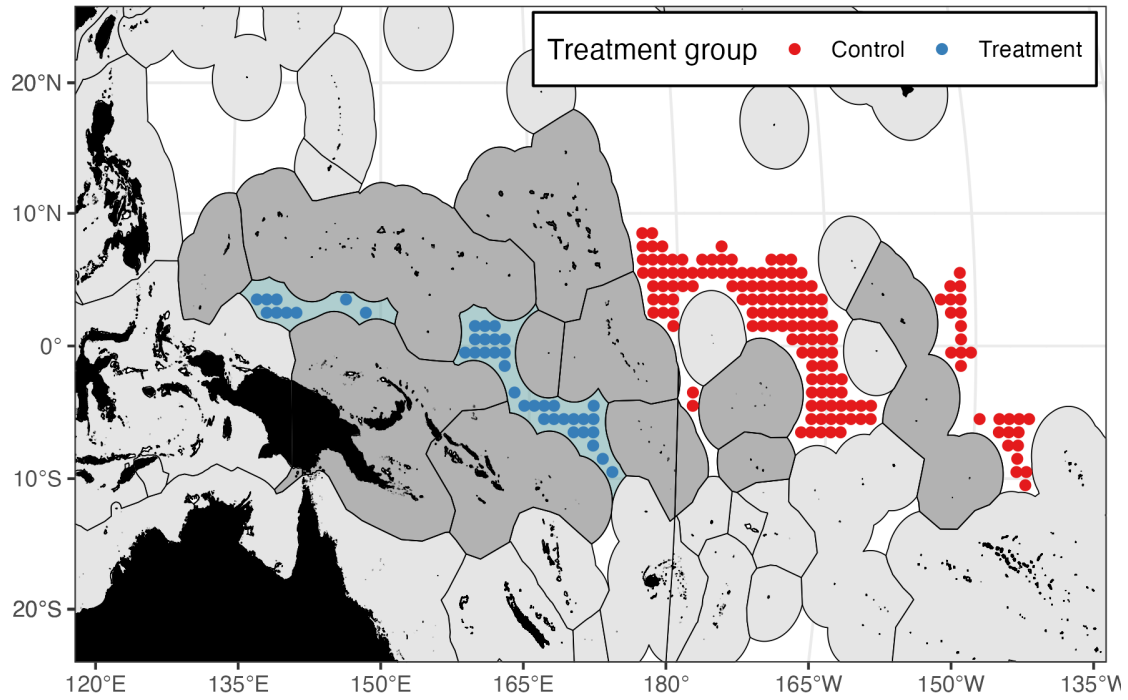


Figure A3: Map of location of treatment and control cells used in our test of changes in purse seine fishing effort in the high seas pockets.

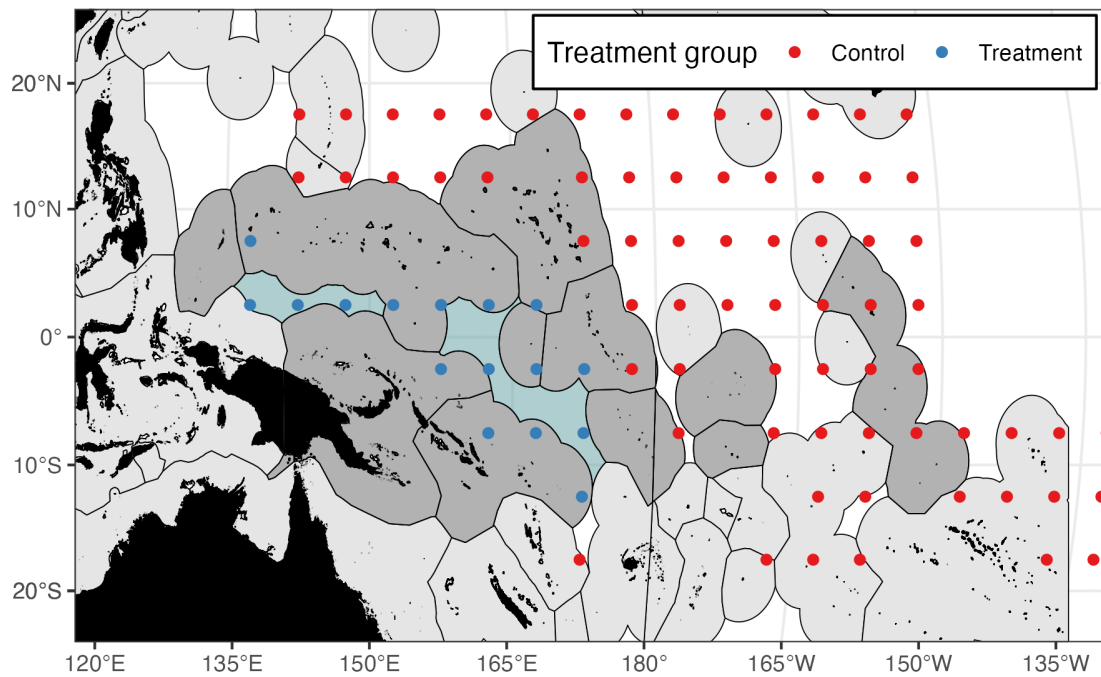


Figure A4: Map of location of treatment and control grid cells used in our test of changes in CPUE for the longline fleet operating in the high seas pockets.

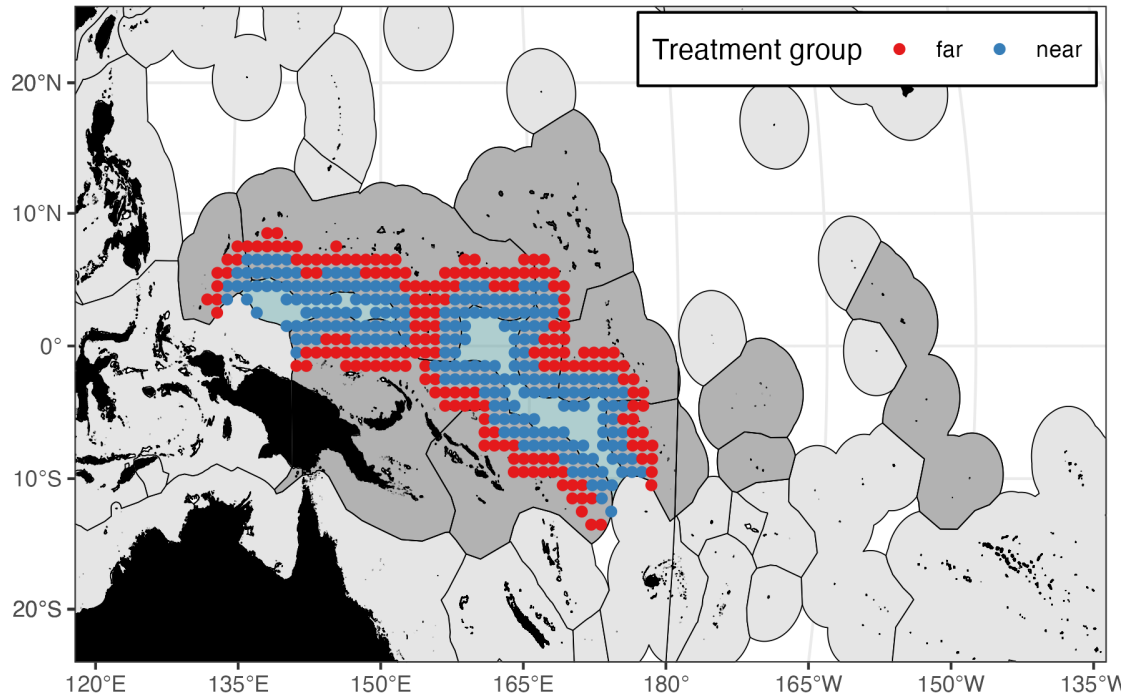


Figure A5: Map of location of treatment and control grid cells used in our test of a spillover benefit to the purse seine fleet.

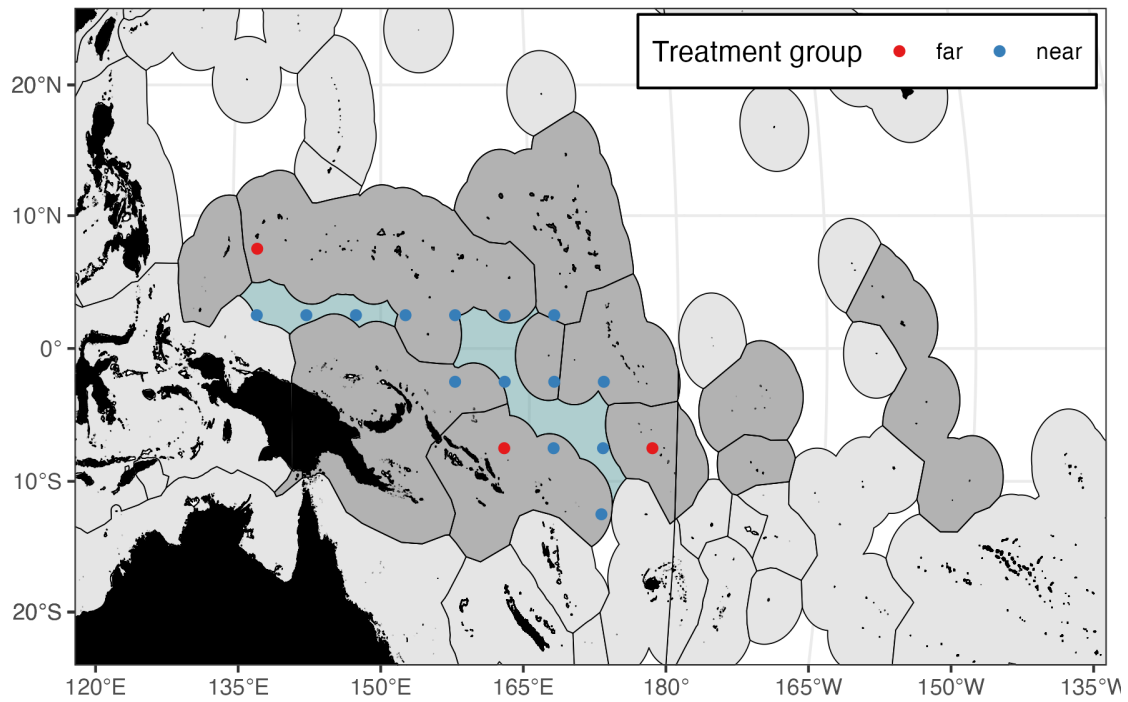


Figure A6: Map of location of treatment and control grid cells used in our test of a spillover benefit to the longline fleet.

## B Additional Results

### B.1 Decrease in purse seine effort in high seas pockets

The reduction in purse seine fishing effort in the high seas pockets is robust across model specifications and outcome transformations. A within-treated specification, which estimates changes relative to the 2009 baseline without a control comparison, shows significant reductions of 54.55 days and 44.02 sets per grid cell per year ( $p < 0.01$ ; Table B1), with event-study estimates confirming no pre-treatment trends (Figure B1). Results comparing trends with other high seas areas within the WCPFC convention area also remain significant when outcomes are transformed using the inverse hyperbolic sine (Table B2; Figure B2).

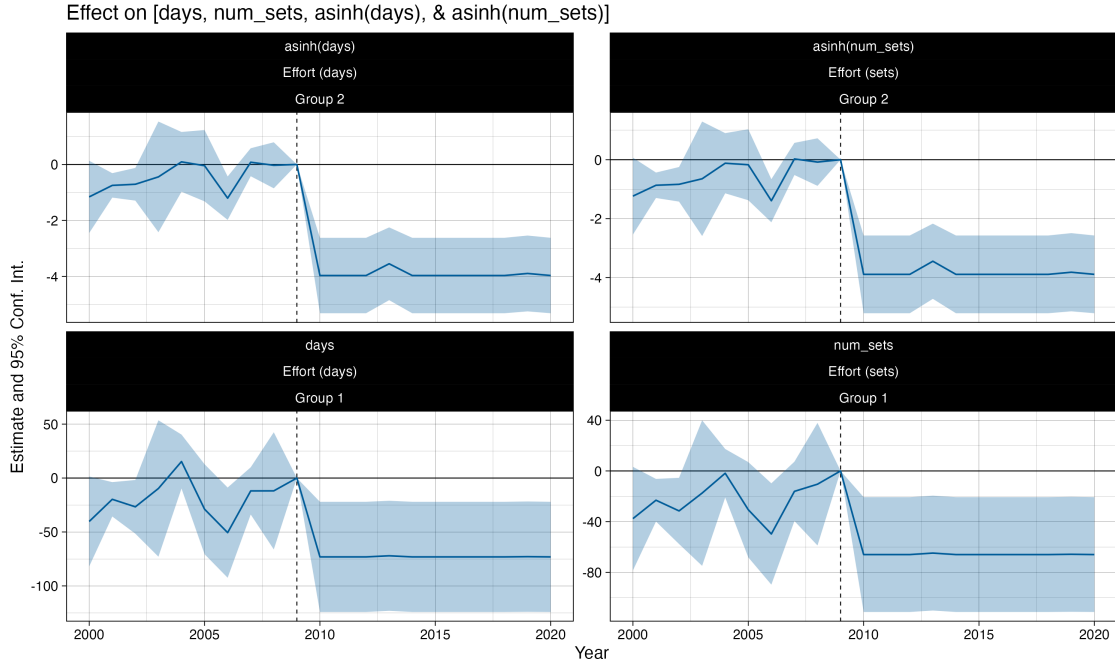


Figure B1: Event-study estimates of the change in purse seine fishing effort within the high seas pockets, estimated using only treated grid cells relative to the 2009 baseline (without a control comparison). Estimates are shown for fishing effort measured in days (left panels) and number of sets (right panels), under the inverse hyperbolic sine (IHS) transformation (top panels) and in levels (bottom panels).

Table B1: Coefficient estimates for change in fishing effort inside the high seas pockets after the closure. Panel A presents results in levels. Panel B presents results in which the dependent variable is transformed using the inverse hyperbolic sine (IHS) transformation.

|                                                  | Effort (days)          | Effort (sets)          |
|--------------------------------------------------|------------------------|------------------------|
| <i>A) Levels</i>                                 |                        |                        |
| Post                                             | -54.555***<br>(13.639) | -44.018***<br>(10.394) |
| R2 Adj.                                          | 0.363                  | 0.332                  |
| <i>B) Inverse-hyperbolic sine transformation</i> |                        |                        |
| Post                                             | -3.505***<br>(0.452)   | -3.310***<br>(0.424)   |
| R2 Adj.                                          | 0.675                  | 0.662                  |
| Num.Obs.                                         | 840                    | 840                    |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. Numbers in parentheses are Conley standard errors using a 220km radius.



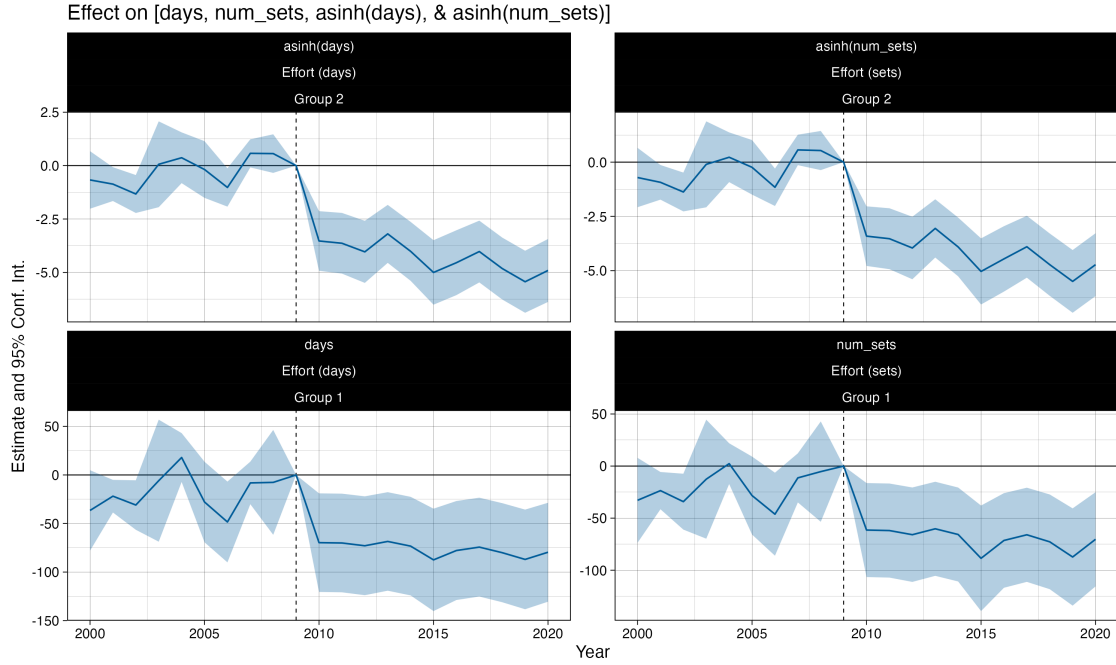


Figure B2: Event-study estimates of the change in purse seine fishing effort in the high seas pockets, relative to fishing effort in comparable high seas regions in the WCPFC convention area. Estimates are shown for fishing effort measured in days (left panels) and number of sets (right panels), under the inverse hyperbolic sine (IHS) transformation (top panels) and in levels (bottom panels). The bottom panels are equivalent to the main text results shown in Figure 2.

Table B2: Coefficient estimates for change in fishing effort inside the high seas pockets after the closure, relative to changes in fishing effort observed in other comparable high seas areas of the WCPFC convention area. Panel A presents results in levels (identical to the main-text estimates in Table 1). Panel B presents results in which the dependent variable is transformed using the inverse hyperbolic sine (IHS) transformation.

|                                                  | Effort (days)          | Effort (sets)          |
|--------------------------------------------------|------------------------|------------------------|
| <i>A) Levels</i>                                 |                        |                        |
| Post x Treated                                   | -59.470***<br>(13.762) | -50.933***<br>(10.607) |
| R2 Adj.                                          | 0.388                  | 0.301                  |
| <i>B) Inverse-hyperbolic sine transformation</i> |                        |                        |
| Post x Treated                                   | -4.036***<br>(0.478)   | -3.881***<br>(0.451)   |
| R2 Adj.                                          | 0.465                  | 0.451                  |
| Num.Obs.                                         | 4011                   | 4011                   |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors using a 220km radius.

## B.2 Change in longline CPUE in high seas pockets

Here we present results for changes in longline CPUE within the high seas pockets for all species and additional model specifications. Across all species combined (Table B3; Figure B3) and for albacore tuna (Table B4; Figure B4), we detect no significant change in longline CPUE. The bigeye tuna results shown in the main text are qualitatively similar under log-transformation, though we note a marginally significant decline in CPUE when measured in metric tons (Table B5; Figure B5). For yellowfin tuna, we estimate a significant increase in log-transformed CPUE measured in number of fish ( $p < 0.05$ ; Table B6; Figure B6).

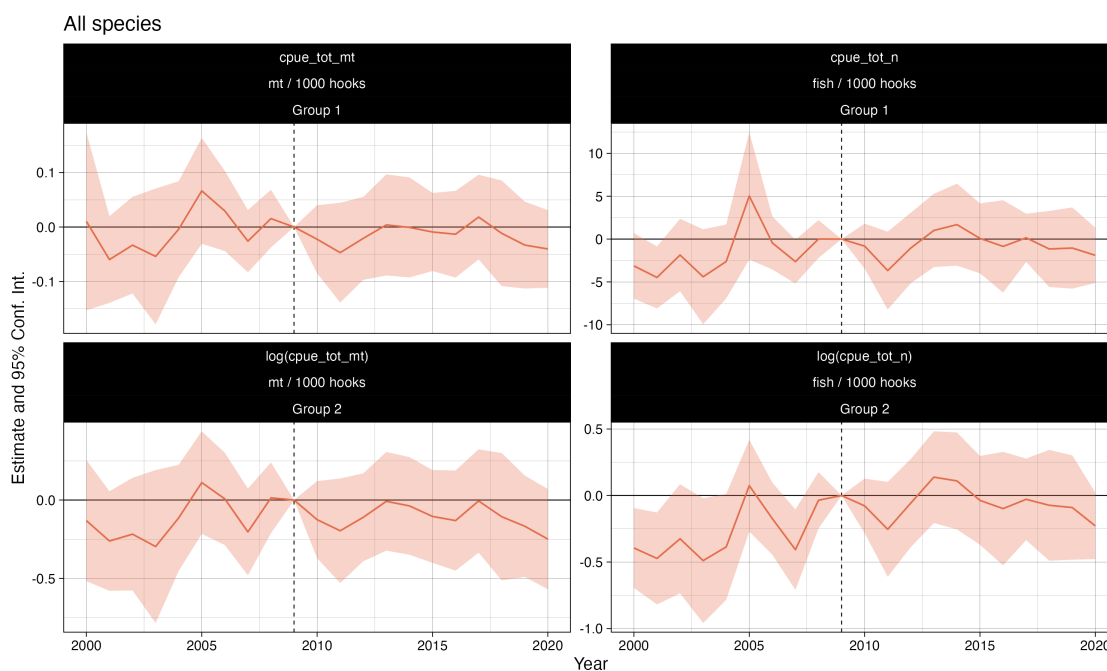


Figure B3: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) across all species combined in the high seas pockets, relative to CPUE in other tropical (20°S to 20°N) high seas regions in the WCPFC convention area. Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B3: Coefficient estimates for change in all tuna CPUE caught by the longline fleet in the high seas pockets after the closure, relative to changes in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

|                           | fish / 1000 hooks | mt / 1000 hooks   |
|---------------------------|-------------------|-------------------|
| <i>A) Levels</i>          |                   |                   |
| Post x Treated            | 0.664<br>(1.605)  | -0.009<br>(0.028) |
| R2 Adj.                   | 0.637             | 0.507             |
| <i>B) Log-transformed</i> |                   |                   |
| Post x Treated            | 0.187<br>(0.139)  | -0.003<br>(0.089) |
| R2 Adj.                   | 0.700             | 0.616             |
| Num.Obs.                  | 1632              | 1632              |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius.

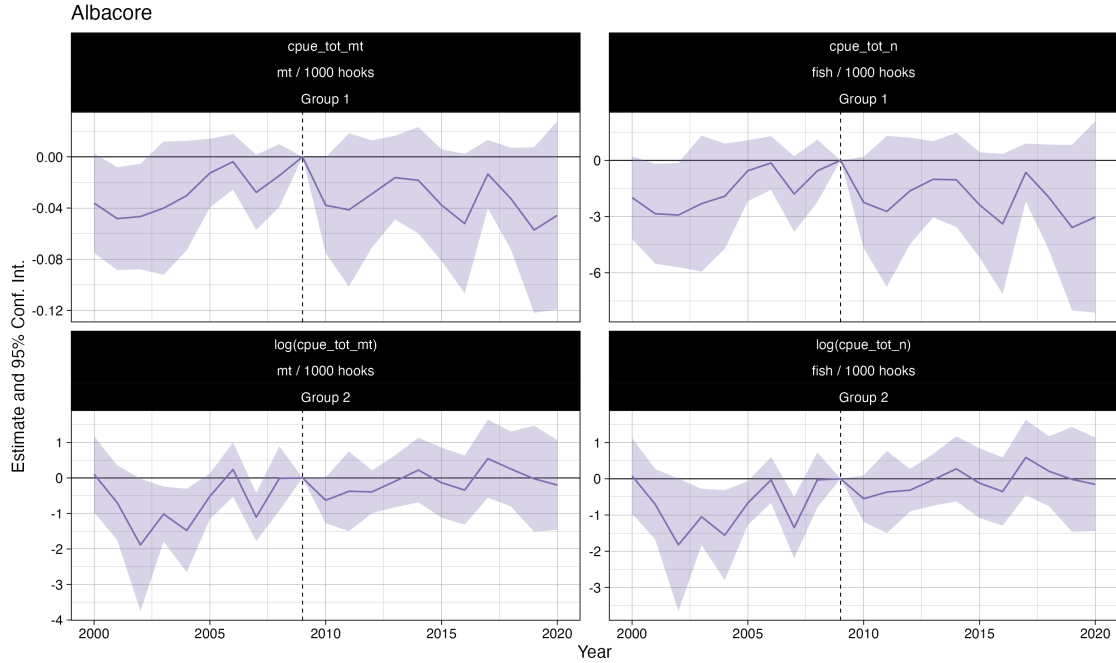


Figure B4: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for albacore tuna in the high seas pockets, relative to CPUE in other tropical (20°S to 20°N) high seas regions in the WCPFC convention area. Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B4: Coefficient estimates for change in albacore tuna CPUE caught by the longline fleet in the high seas pockets after the closure, relative to changes in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

|                           | fish / 1000 hooks | mt / 1000 hooks   |
|---------------------------|-------------------|-------------------|
| <i>A) Levels</i>          |                   |                   |
| Post x Treated            | -0.645<br>(0.793) | -0.009<br>(0.012) |
| Num.Obs.                  | 1606              | 1606              |
| R2 Adj.                   | 0.856             | 0.852             |
| <i>B) Log-transformed</i> |                   |                   |
| Post x Treated            | 0.644<br>(0.498)  | 0.540<br>(0.485)  |
| Num.Obs.                  | 1606              | 1603              |
| R2 Adj.                   | 0.821             | 0.809             |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius.

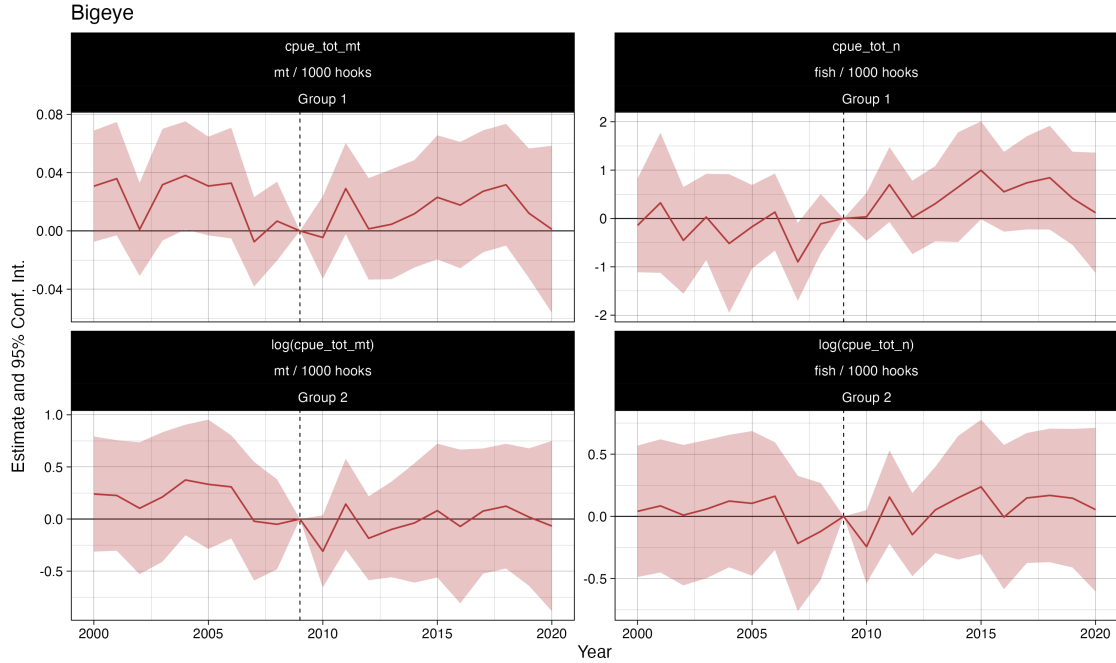


Figure B5: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for bigeye tuna in the high seas pockets, relative to CPUE in other tropical (20°S to 20°N) high seas regions in the WCPFC convention area. Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels). The top panels are equivalent to the main text results shown in Figure 3.

Table B5: Coefficient estimates for change in bigeye tuna CPUE caught by the longline fleet in the high seas pockets after the closure, relative to changes in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area. Panel A presents results in levels (same as in Table 2). Panel B presents results in which the dependent variable is log-transformed.

|                           | fish / 1000 hooks | mt / 1000 hooks    |
|---------------------------|-------------------|--------------------|
| <i>A) Levels</i>          |                   |                    |
| Post x Treated            | 0.697*<br>(0.363) | -0.003<br>(0.011)  |
| R2 Adj.                   | 0.798             | 0.829              |
| <i>B) Log-transformed</i> |                   |                    |
| Post x Treated            | 0.055<br>(0.106)  | -0.176*<br>(0.094) |
| R2 Adj.                   | 0.852             | 0.880              |
| Num.Obs.                  | 1631              | 1631               |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius.



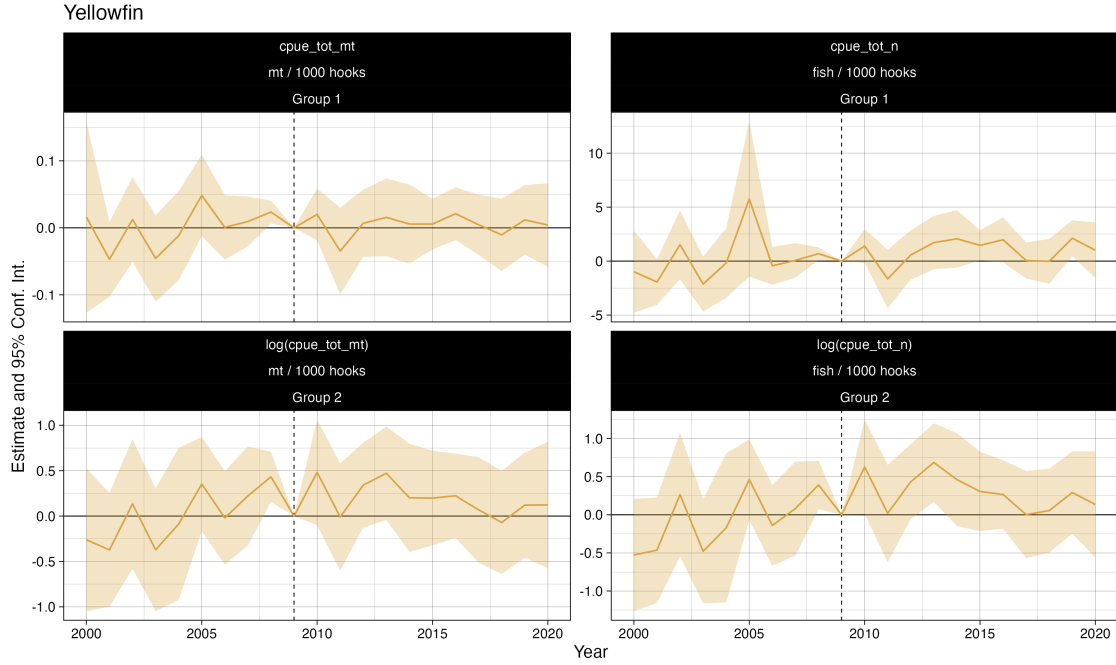


Figure B6: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for yellowfin tuna in the high seas pockets, relative to CPUE in other tropical (20°S to 20°N) high seas regions in the WCPFC convention area. Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B6: Coefficient estimates for change in yellowfin tuna CPUE caught by the longline fleet in the high seas pockets after the closure, relative to changes in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

|                           | fish / 1000 hooks  | mt / 1000 hooks  |
|---------------------------|--------------------|------------------|
| <i>A) Levels</i>          |                    |                  |
| Post x Treated            | 0.598<br>(0.857)   | 0.002<br>(0.016) |
| R2 Adj.                   | 0.538              | 0.607            |
| <i>B) Log-transformed</i> |                    |                  |
| Post x Treated            | 0.329**<br>(0.151) | 0.171<br>(0.117) |
| R2 Adj.                   | 0.702              | 0.704            |
| Num.Obs.                  | 1632               | 1632             |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius.

### B.3 Spillover benefits to fisheries adjacent to high seas pockets

**Purse seine.** We find no evidence of spillover benefits to the purse seine fleet operating near the high seas pockets for any species or model specification. Results are null across all species combined (Table B7; Figure B7), bigeye tuna (Table B8; Figure B8), skipjack tuna (Table B9; Figure B9), and yellowfin tuna (Table B10; Figure B10), in both levels and log-transformed specifications.

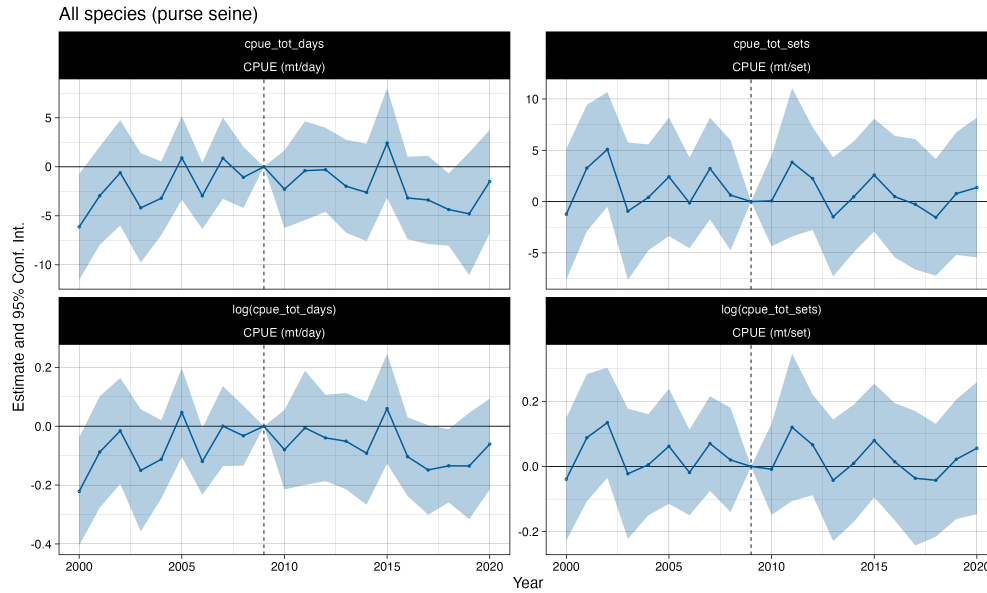


Figure B7: Event-study estimates of the change in purse seine catch-per-unit-effort (CPUE) across all species combined within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Both the near (treated) and far (control) samples are restricted to grid cells within PNA member Exclusive Economic Zones. Estimates are shown for CPUE measured in tons per day (left panels) and tons per set (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B7: Coefficient estimates for change in all tuna CPUE caught by the purse seine fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles and inside PNA nation's Exclusive Economic Zones. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

|                           | CPUE (mt/day)     | CPUE (mt/set)     |
|---------------------------|-------------------|-------------------|
| <i>A) Levels</i>          |                   |                   |
| Post x Near               | -0.294<br>(0.827) | -0.468<br>(1.050) |
| R2 Adj.                   | 0.273             | 0.209             |
| <i>B) Log-transformed</i> |                   |                   |
| Post x Near               | -0.007<br>(0.028) | -0.007<br>(0.034) |
| R2 Adj.                   | 0.258             | 0.221             |
| Num.Obs.                  | 5960              | 5960              |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

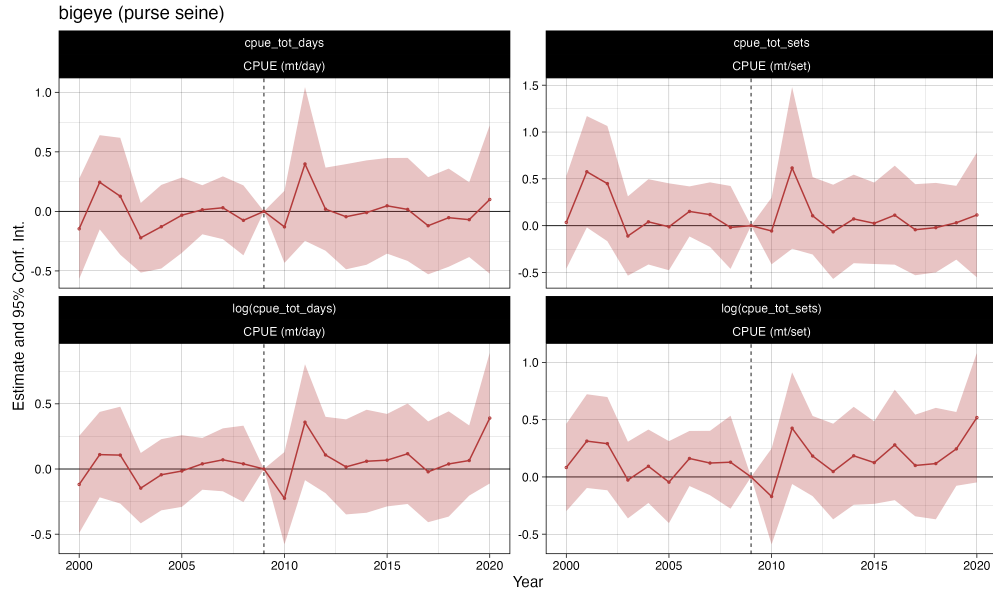


Figure B8: Event-study estimates of the change in purse seine catch-per-unit-effort (CPUE) for bigeye tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Both the near (treated) and far (control) samples are restricted to grid cells within PNA member Exclusive Economic Zones. Estimates are shown for CPUE measured in tons per day (left panels) and tons per set (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B8: Coefficient estimates for change in big-eye tuna CPUE caught by the purse seine fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles and inside PNA nation's Exclusive Economic Zones. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

|                           | CPUE (mt/day)    | CPUE (mt/set)     |
|---------------------------|------------------|-------------------|
| <i>A) Levels</i>          |                  |                   |
| Post x Near               | 0.055<br>(0.104) | -0.005<br>(0.121) |
| R2 Adj.                   | 0.174            | 0.209             |
| <i>B) Log-transformed</i> |                  |                   |
| Post x Near               | 0.086<br>(0.086) | 0.081<br>(0.096)  |
| R2 Adj.                   | 0.227            | 0.279             |
| Num.Obs.                  | 5908             | 5908              |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

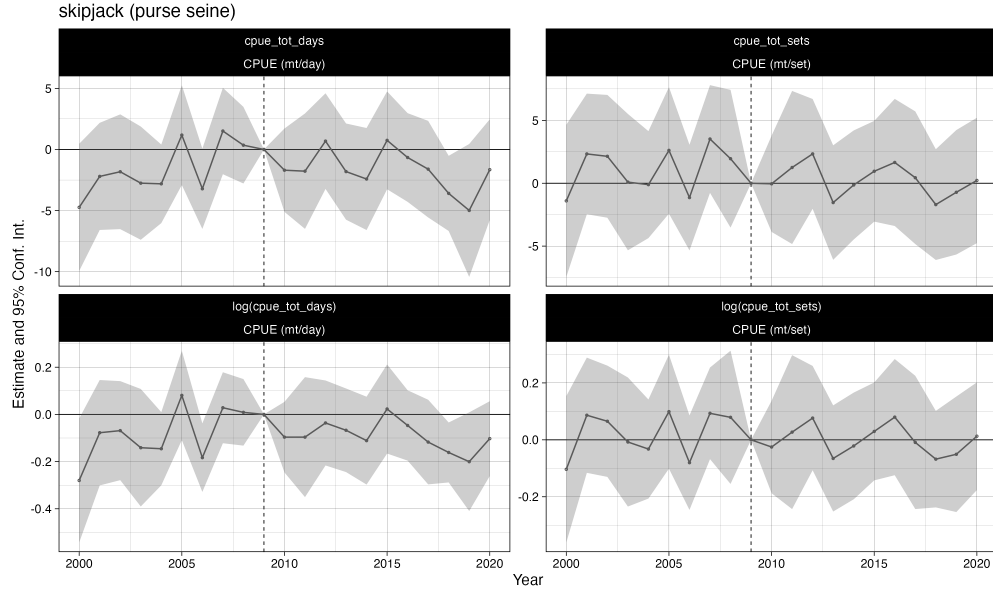


Figure B9: Event-study estimates of the change in purse seine catch-per-unit-effort (CPUE) for skipjack tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Both the near (treated) and far (control) samples are restricted to grid cells within PNA member Exclusive Economic Zones. Estimates are shown for CPUE measured in tons per day (left panels) and tons per set (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels). The top panels are equivalent to the main text results shown in Figure 4.

Table B9: Coefficient estimates for change in skipjack tuna CPUE caught by the purse seine fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles and inside PNA nation's Exclusive Economic Zones. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

|                           | CPUE (mt/day)     | CPUE (mt/set)     |
|---------------------------|-------------------|-------------------|
| <i>A) Levels</i>          |                   |                   |
| Post x Near               | -0.496<br>(0.679) | -0.865<br>(0.818) |
| R2 Adj.                   | 0.333             | 0.245             |
| <i>B) Log-transformed</i> |                   |                   |
| Post x Near               | -0.024<br>(0.031) | -0.026<br>(0.038) |
| R2 Adj.                   | 0.316             | 0.244             |
| Num.Obs.                  | 5960              | 5960              |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.



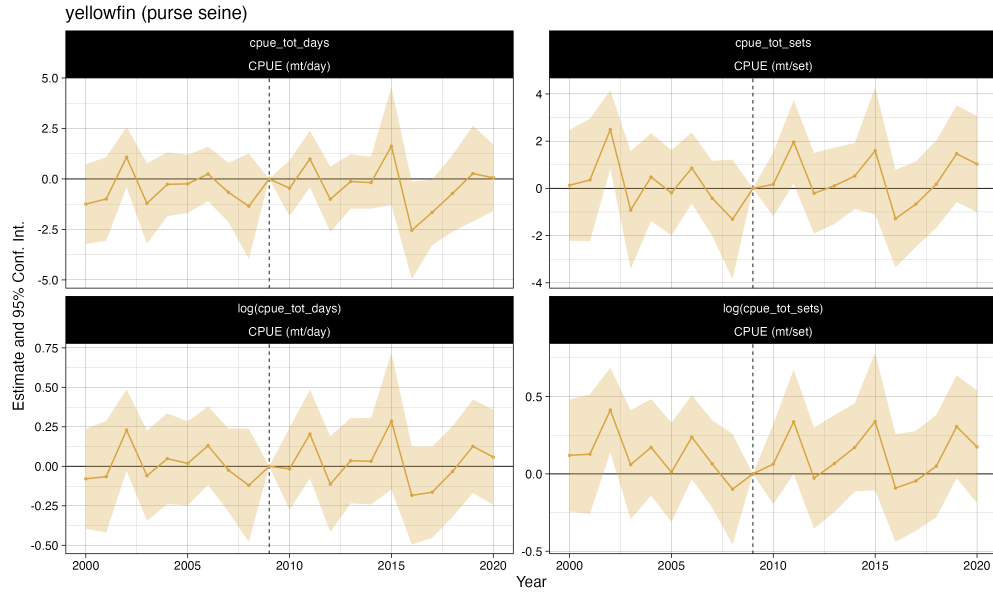


Figure B10: Event-study estimates of the change in purse seine catch-per-unit-effort (CPUE) for yellowfin tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Both the near (treated) and far (control) samples are restricted to grid cells within PNA member Exclusive Economic Zones. Estimates are shown for CPUE measured in tons per day (left panels) and tons per set (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B10: Coefficient estimates for change in yellowfin tuna CPUE caught by the purse seine fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles and inside PNA nation's Exclusive Economic Zones. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

|                           | CPUE (mt/day)    | CPUE (mt/set)    |
|---------------------------|------------------|------------------|
| <i>A) Levels</i>          |                  |                  |
| Post x Near               | 0.146<br>(0.281) | 0.402<br>(0.300) |
| R2 Adj.                   | 0.221            | 0.280            |
| <i>B) Log-transformed</i> |                  |                  |
| Post x Near               | 0.015<br>(0.037) | 0.022<br>(0.044) |
| R2 Adj.                   | 0.244            | 0.304            |
| Num.Obs.                  | 5956             | 5956             |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

576 **Longline.** There are only 3 cells within 100–200 nautical miles, leaving us with a reduced control  
577 group for this portion of the analyses. These results are presented to comply with our prereg-  
578 istration, but we note the lack of robustness that comes with such a limited sample size. With  
579 this caveat in mind, we detect no significant spillover effects for all species combined (Table B11;  
580 Figure B11), bigeye tuna (Table B13; Figure B13), or yellowfin tuna (Table B14; Figure B14).  
581 The only marginally significant result is for albacore tuna under the log-transformed specification  
582 ( $p < 0.1$ ; Table B12; Figure B12).

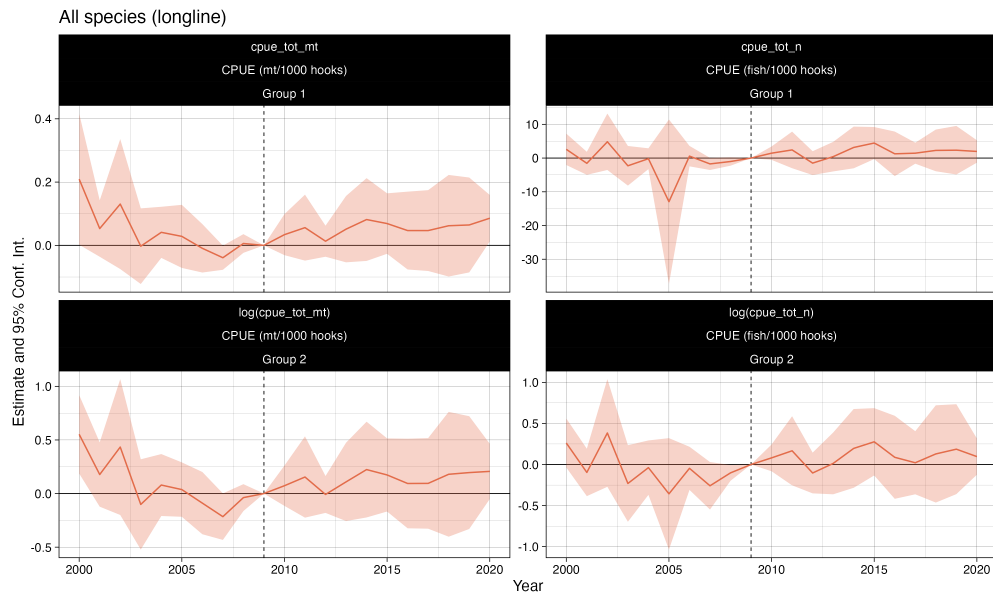


Figure B11: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) across all species combined within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B11: Coefficient estimates for change in all tuna CPUE caught by the longline fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

|                           | CPUE (fish/1000 hooks) | CPUE (mt/1000 hooks) |
|---------------------------|------------------------|----------------------|
| <i>A) Levels</i>          |                        |                      |
| Post x Near               | 2.983<br>(2.918)       | 0.015<br>(0.039)     |
| R2 Adj.                   | 0.578                  | 0.360                |
| <i>B) Log-transformed</i> |                        |                      |
| Post x Near               | 0.145<br>(0.174)       | 0.057<br>(0.128)     |
| R2 Adj.                   | 0.662                  | 0.404                |
| Num.Obs.                  | 347                    | 347                  |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.



Figure B12: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for albacore tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B12: Coefficient estimates for change in albacore tuna CPUE caught by the longline fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

|                           | CPUE (fish/1000 hooks) | CPUE (mt/1000 hooks) |
|---------------------------|------------------------|----------------------|
| <i>A) Levels</i>          |                        |                      |
| Post x Near               | -0.487<br>(1.260)      | -0.011<br>(0.019)    |
| R2 Adj.                   | 0.878                  | 0.881                |
| <i>B) Log-transformed</i> |                        |                      |
| Post x Near               | 1.326*<br>(0.724)      | 1.403*<br>(0.810)    |
| R2 Adj.                   | 0.843                  | 0.819                |
| Num.Obs.                  | 339                    | 339                  |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

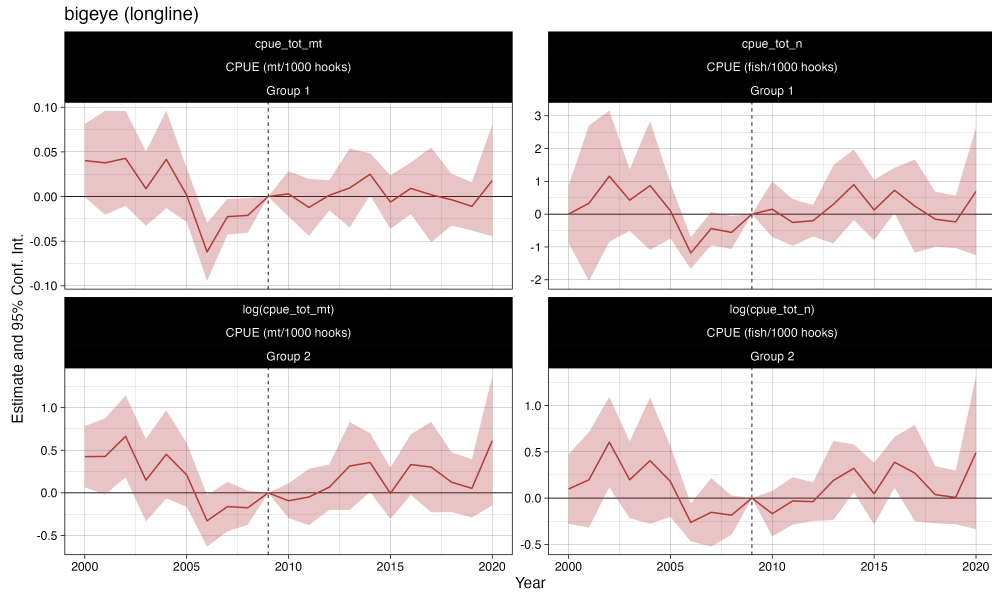


Figure B13: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for bigeye tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B13: Coefficient estimates for change in bigeye tuna CPUE caught by the longline fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

|                           | CPUE (fish/1000 hooks) | CPUE (mt/1000 hooks) |
|---------------------------|------------------------|----------------------|
| <i>A) Levels</i>          |                        |                      |
| Post x Near               | 0.136<br>(0.433)       | 0.000<br>(0.013)     |
| R2 Adj.                   | 0.683                  | 0.709                |
| <i>B) Log-transformed</i> |                        |                      |
| Post x Near               | 0.028<br>(0.134)       | 0.032<br>(0.152)     |
| R2 Adj.                   | 0.812                  | 0.848                |
| Num.Obs.                  | 347                    | 347                  |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

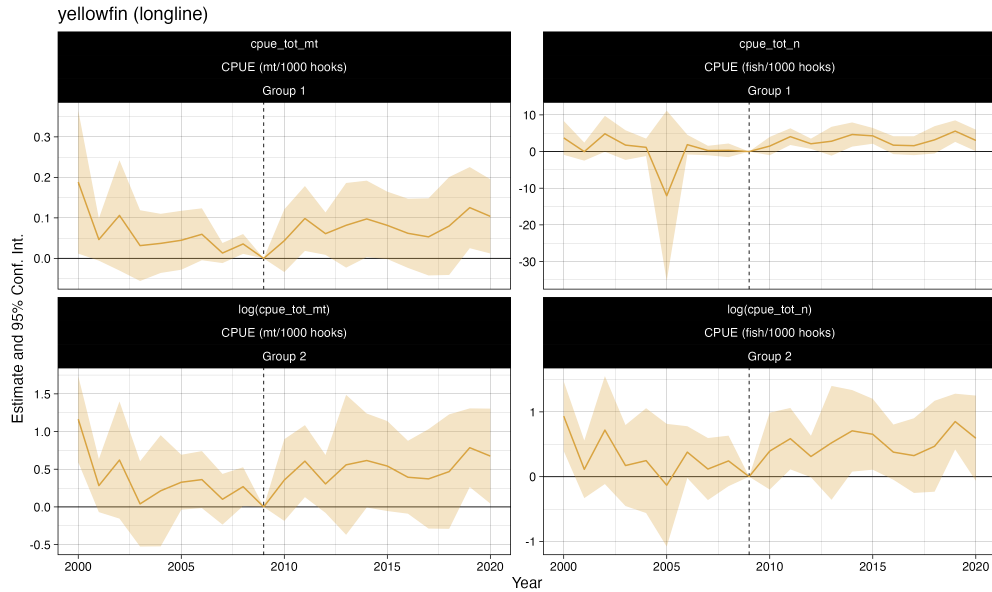


Figure B14: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for yellowfin tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B14: Coefficient estimates for change in yellowfin tuna CPUE caught by the longline fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

|                           | CPUE (fish/1000 hooks) | CPUE (mt/1000 hooks) |
|---------------------------|------------------------|----------------------|
| <i>A) Levels</i>          |                        |                      |
| Post x Near               | 3.346<br>(2.147)       | 0.026<br>(0.030)     |
| R2 Adj.                   | 0.383                  | 0.382                |
| <i>B) Log-transformed</i> |                        |                      |
| Post x Near               | 0.254<br>(0.209)       | 0.176<br>(0.200)     |
| R2 Adj.                   | 0.518                  | 0.493                |
| Num.Obs.                  | 347                    | 347                  |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

## C Robustness Tests

### C.1 Results by high seas pocket

To assess whether our main results are driven by a single pocket, and as a robustness test to account for the exemption granted to the Philippines, we re-estimate the main specifications for bigeye tuna longline CPUE (Table 2) and skipjack tuna purse seine CPUE spillover (Table 3) separately for each high seas pocket. For the longline CPUE analysis, we drop treated grid cells from one pocket at a time while retaining the full set of control cells. For the spillover analysis, we split the entire sample—both near and far grid cells—east and west of 152.5°. The increase in bigeye CPUE measured in number of fish is significant for both the western ( $p < 0.01$ ) and eastern ( $p < 0.05$ ) pockets, with a larger point estimate in the western pocket (Table C1). The null spillover result for skipjack tuna holds for both pockets individually (Table C2).

Table C1: Coefficient estimates for change in bigeye tuna CPUE caught by the longline fleet in the high seas pockets after the closure, relative to changes in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area. Models are estimated separately for HSP1 (western pocket,  $\text{lon} \leq 152.5^\circ$ ) and HSP2 (eastern pocket,  $\text{lon} > 152.5^\circ$ ; See Figure 1). The first column shows our main text estimates as in Table 2.

|                                    | sample: Full sample | sample: HSP1        | sample: HSP2       |
|------------------------------------|---------------------|---------------------|--------------------|
| <i>A) CPUE (fish / 1000 hooks)</i> |                     |                     |                    |
| Post x Treated                     | 0.697*<br>(0.363)   | 1.537***<br>(0.520) | 0.397**<br>(0.196) |
| R2 Adj.                            | 0.798               | 0.801               | 0.804              |
| <i>B) CPUE (mt / 1000 hooks)</i>   |                     |                     |                    |
| Post x Treated                     | -0.003<br>(0.011)   | 0.020<br>(0.013)    | -0.011<br>(0.010)  |
| R2 Adj.                            | 0.829               | 0.830               | 0.835              |
| Num.Obs.                           | 1631                | 1407                | 1529               |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius.

Table C2: Coefficient estimates for change in skipjack tuna CPUE caught by the purse seine fleet in areas within 100 nautical miles of each high seas pocket after the closure. Models are estimated separately for HSP1 (western pocket,  $\text{lon} \leq 152.5^\circ$ ) and HSP2 (eastern pocket,  $\text{lon} > 152.5^\circ$ ; See Figure 1). The first column shows our main text estimates as in Table 3.

|                         | sample: Full sample | sample: HSP1      | sample: HSP2      |
|-------------------------|---------------------|-------------------|-------------------|
| <i>A) CPUE (mt/day)</i> |                     |                   |                   |
| Post x Near             | -0.496<br>(0.679)   | -0.132<br>(0.635) | -0.952<br>(0.957) |
| R2 Adj.                 | 0.333               | 0.433             | 0.319             |
| <i>B) CPUE (mt/set)</i> |                     |                   |                   |
| Post x Near             | -0.865<br>(0.818)   | -0.925<br>(0.715) | -1.206<br>(1.141) |
| R2 Adj.                 | 0.245               | 0.353             | 0.213             |
| Num.Obs.                | 5960                | 1645              | 4315              |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.